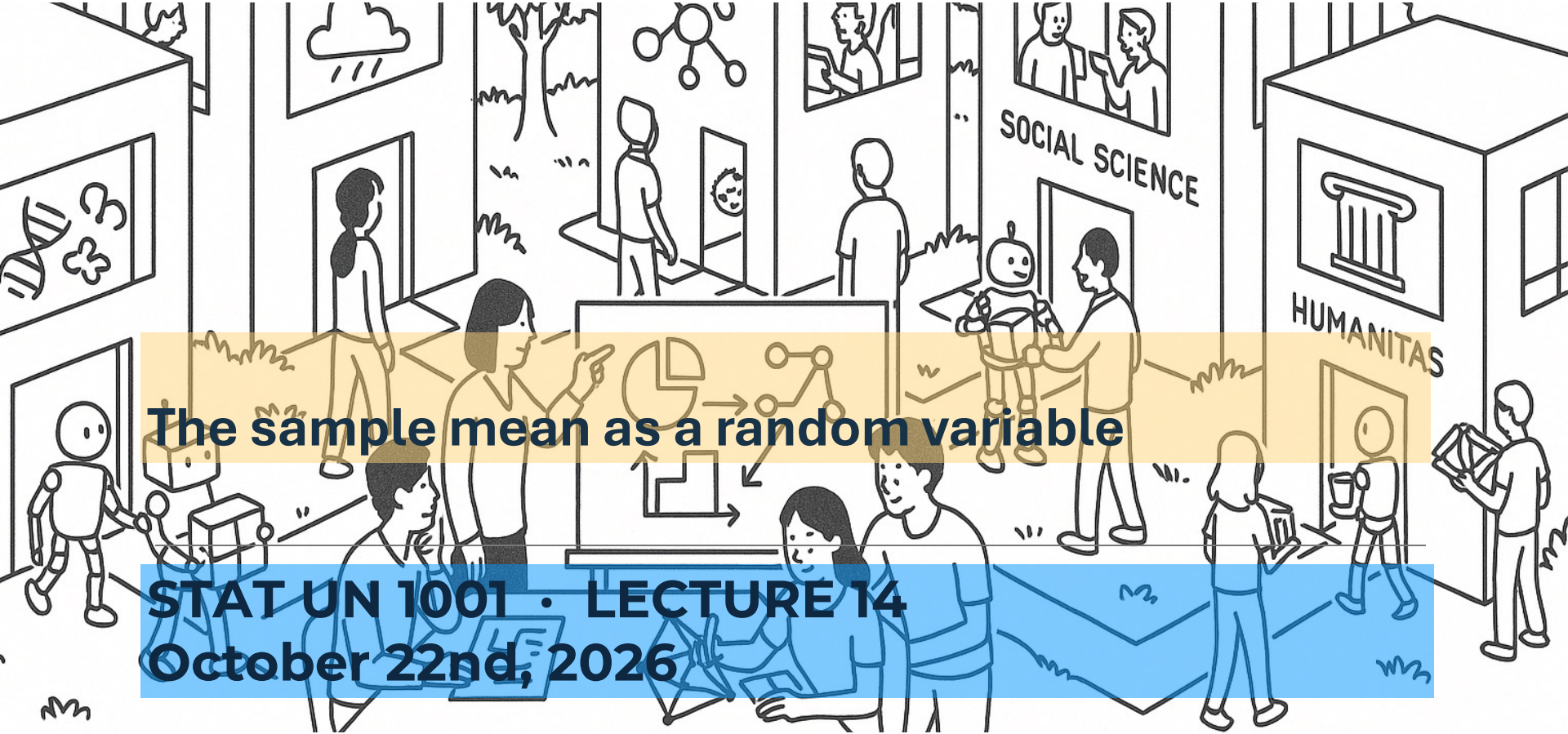


The sample mean as a random variable

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A sample mean is a random variable

The number changes because the sample is selected by chance.

Object	What it means
Population parameter μ	A fixed (usually unknown) feature of the population
Chance process	Select a random sample of n units
Outcome: Statistic $\bar{x}$	Compute the average of the selected values
Random variable	$\bar{x}$ can take different values in different samples

Before the sample is selected, $\bar{x}$ is a possible numerical outcome. After the sample is selected, it is one observed statistic.

Name the chance process before calculating

Example: measure a numerical value for a random sample of 20 households.

Part of the setup	Answer
Chance process	Select a simple random sample of 20 households
One outcome	The particular 20 households selected
Sample space	All possible samples of 20 households
Random variable	$\bar{X}$ = the average number of household members in the selected sample
Event	For example, $\bar{X} \geq 2$

The event is a set of possible sample outcomes—not just one sample's value.

Build a SIMPLE sampling distribution

Population values: 2, 4, 6. Select 2 without replacement.
Each unordered sample is equally likely.

Possible sample	Sample mean $\bar{X}$	Probability
{2, 4}	3	1/3
{2, 6}	4	1/3
{4, 6}	5	1/3

The sampling distribution of $\bar{X}$ is the list of possible sample means and their probabilities.

Study an event for the sample mean

Event: the selected pair has an average of at least 4

Possible $\bar{X}$	Probability	Counts for $\bar{X} \geq 4$?
3	1/3	No
4	1/3	Yes
5	1/3	Yes

$$P(\bar{X} \geq 4)$$
$$= P(\bar{X} = 4) + P(\bar{X} = 5)$$
$$= 1/3 + 1/3 = 2/3$$

In words: over repeated random samples of two, about two-thirds would have a mean of at least 4 in this population.

Standard error = the spread of a statistic

Quantity	What its spread describes
SD of individual values	How much units differ within one population or sample
SE($\bar{X}$)	How much sample means differ across repeated random samples

SE($\bar{X}$) is the standard deviation of the sampling distribution of $\bar{X}$.

Under the usual model:

$$SE(\bar{X}) = \sigma / \sqrt{n}$$

Interpretation: a typical sample mean is about one standard error from the center of its sampling distribution.

Why larger samples usually improve precision

Suppose the population SD is $\sigma = 8$.

Sample size n	$SE(\bar{X}) = 8 / \sqrt{n}$	What changes?
1	8	One observation: most variable
4	4	Four observations: half the SE
16	2	Sixteen observations: one-quarter the SE

A larger n narrows the sampling distribution around the same center. It improves precision, but it does not automatically remove bias.